

WELCOME TO MY PROJECT

What is MINI-IDS?

MINI-IDS is a lightweight, real-time network intrusion detection system that captures live packets, analyzes traffic for suspicious patterns, and automatically blocks malicious IP addresses to reduce attacker dwell time.

Purpose:

Apply data structures & algorithms in a cybersecurity context; provide hands-on learning and a production-style monitoring tool.

Goals:

Monitor traffic in real time, detect brute-force and protocol-specific attacks, and enforce automated responses.

Problem Statement:

Networks produce massive volumes of traffic; manual monitoring is impractical and many commercial IDS solutions are complex or costly. Attack frequency and sophistication are increasing, creating a need for a compact, effective detection tool.

Existing Issues

- High traffic volumes
- Time-consuming manual analysis
- Complex/expensive commercial IDS
- Rapidly evolving attack techniques

Our Approach

Provide a lightweight system for real-time detection and automated mitigation focused on common brute-force and protocol attacks.

Objectives:

MINI-IDS targets practical learning and operational capability.

Key objectives:

- Capture live network packets across interfaces
- Detect brute-force and protocol-specific attacks (SSH, FTP, RDP)
- Automatically block suspicious IP addresses
- Persistently store detailed attack logs
- Demonstrate DSA concepts through implementation
- Provide efficient search and sorting for log analysis

Scope & Target Users:

Application Areas

- Educational institutions and labs
- Small business and home networks
- Research laboratories for IDS experiments

Target Users

- Cybersecurity students and instructors
- Network administrators
- Security researchers
- IT professionals exploring lightweight IDS

Key Features:

1. Monitoring

Real-time packet capture, live traffic monitoring, multi-interface support.

2. Detection

Protocol-focused detection for SSH (22), FTP (21), and RDP (3389); brute-force pattern recognition.

3. Security

Automatic IP blocking, attack logging, and real-time alerts for rapid response.

4. DSA Integration

Implements linked lists, hash sets, linear & binary search, and classic sorting algorithms for log management and detection efficiency.

System Workflow:

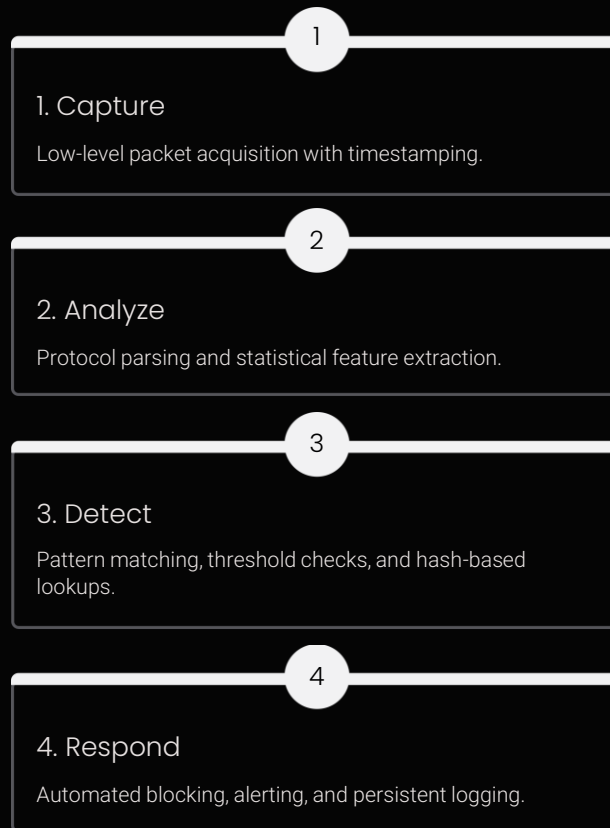


Data flows from capture through analysis to automated response and persistent logging. This pipeline supports near-real-time mitigation while retaining forensic data for later analysis.

System Architecture:

Architecture layers provide separation of concerns for reliability and testability.

- User → Network Interface
- Packet Capture Engine (Npcap/libpcap)
- Packet Analyzer → Attack Detection Module
- Log Storage & Reporting



Technologies & Implementation Details:

Core Components:

- Language: C++ for performance and low-level packet access
- Packet capture: Npcap / libpcap
- IDE: Visual Studio Code
- Platform: Windows and Linux support

Data Structures & Algorithms:

- Linked List: queueing recent connection attempts
- Hash Set: fast IP lookup and blacklist checks
- Search: linear for short lists, binary for sorted logs
- Sorting: bubble and selection used for teaching; more efficient sorts can be plugged in

Advantages & Next Steps:

MINI-IDS is designed to be lightweight, fast, and educational. It provides real-time monitoring, automatic mitigation, and persistent logging while serving as a practical demonstration of data structures and algorithms in cybersecurity.

Benefits

- Low resource footprint and real-time response
- Easy to deploy in labs and small networks
- Cross-platform compatibility and persistent logs for forensics

Future Work

Integrate more efficient sorting/search structures, expand protocol coverage, add ML-assisted anomaly detection, and a web UI for visualization and queryable logs.

Practical Implementation:

1.Access Control:Get Access By Giving Password.

```
M /c/Users/LAPTOP GADGET/Downloads
LAPTOP GADGET@POROSH UCRT64 /c/Users/LAPTOP GADGET/Downloads
# ./ids.exe

-----
ACCESS CONTROL
-----

Password: *****
```

2. CLI – Interface:

```
M /c/Users/LAPTOP GADGET/Downloads

-----
MINI-IDS - A Real Time Data Capturing Tool
REAL-TIME NETWORK INTRUSION DETECTION

WARNING: THIS CAPTURES REAL NETWORK TRAFFIC
RUN AS ADMINISTRATOR ON WINDOWS
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-----
MAIN MENU
-----

[1] LIST NETWORK INTERFACES:
[2] START REAL-TIME CAPTURE:
[3] VIEW STATISTICS:
[4] VIEW BLOCKED IP LIST :
[5] VIEW ATTACK LOGS (LINKED LIST TRAVERSAL):
[6] LINEAR SEARCH IN LOGS :
[7] BINARY SEARCH (SORTED BY IP):
[8] SORTING DEMO (BUBBLE / SELECTION SORT):
[9] EXPORT ALL LOGS TO FILE:
[0] EXIT

Enter your choice: |
```

3.List Network Interface:

```
/c/Users/LAPTOP GADGET/Downloads

=====
MAIN MENU
=====
[1] LIST NETWORK INTERFACES:
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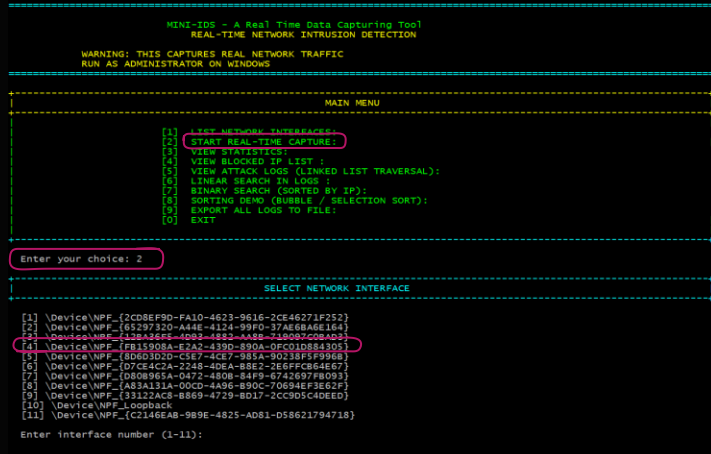
Enter your choice: 1

=====
AVAILABLE NETWORK INTERFACES
=====

[1] \Device\NPF_{2CD8EF9D-FA10-4623-9616-2CE46271F252} --> WAN Miniport (IPv6)
[2] \Device\NPF_{65297320-A44E-4124-99F0-37AE6BA6E164} --> WAN Miniport (IP)
[3] \Device\NPF_{12BA36F5-4D93-4882-AA8B-719097C0BAD3} --> WAN Miniport (Network Monitor)
[4] \Device\NPF_{FB15908A-E2A2-439D-890A-0FC01D884305} --> Intel(R) Wi-Fi 6 AX200 160MHz
[5] \Device\NPF_{8D6D3D2D-C5E7-4CE7-985A-90238F5F996B} --> VMware Virtual Ethernet Adapter for VMnet8
[6] \Device\NPF_{D7CE4C2A-2248-4DEA-B8E2-2E6FFCB64E67} --> VMware Virtual Ethernet Adapter for VMnet1
[7] \Device\NPF_{D80B965A-0472-480B-84F9-6742697FB093} --> Microsoft Wi-Fi Direct Virtual Adapter #2
[8] \Device\NPF_{A83A131A-00CD-4A96-B90C-70694EF3E62F} --> Microsoft Wi-Fi Direct Virtual Adapter
[9] \Device\NPF_{33122AC8-B869-4729-BD17-2CC9D5C4DEED} --> TAP-Windows Adapter V9
[10] \Device\NPF_{Loopback} --> Adapter for loopback traffic capture
[11] \Device\NPF_{C2146EAB-9B9E-4825-AD81-D58621794718} --> Realtek PCIe GbE Family Controller

Press any key to continue...|
```

4. Real-Time Packet Capture:



C:\Users\T\LAPTOP GADGET\Downloads

#	IP	Port	Protocol
76177	172.17.113.172	224.0.0.251	UDP
76178	172.17.113.172	224.0.0.252	UDP
76179	172.17.113.172	224.0.0.251	UDP
76180	172.17.113.172	224.0.0.251	UDP
76181	172.17.113.172	224.0.0.251	UDP
76182	172.17.113.172	224.0.0.251	UDP
76183	172.17.112.241	57.144.152.141	TCP
76184	57.144.152.141	172.17.112.241	TCP
76185	57.144.152.141	172.17.112.241	TCP
76186	172.17.112.241	57.144.152.141	TCP
76187	172.17.113.23	224.0.0.252	UDP
76188	172.17.112.241	172.17.112.241	TCP
76189	172.17.112.254	172.17.112.241	UDP
76190	172.17.112.241	2.19.212.173	TCP
76191	2.19.212.173	172.17.112.241	TCP
76192	172.17.112.241	2.19.212.173	TCP
76193	172.17.112.241	2.19.212.173	TCP
76194	2.19.212.173	172.17.112.241	TCP
76195	2.19.212.173	172.17.112.241	TCP
76196	172.17.112.241	2.19.212.173	TCP
76197	172.17.112.241	2.19.212.173	TCP
76198	152.42.174.114	172.17.112.241	TCP
76199	172.17.112.241	152.42.174.114	TCP
76200	152.42.174.114	172.17.112.241	TCP
76201	3.33.193.183	172.17.112.241	TCP
76202	172.17.112.241	3.33.193.183	TCP
76203	3.33.193.183	172.17.112.241	TCP
76204	172.17.113.23	224.0.0.252	UDP
76205	172.17.112.241	2.19.212.173	TCP
76206	2.19.212.173	172.17.112.241	TCP
76207	172.17.112.241	4.213.25.241	TCP
76208	4.213.25.241	172.17.112.241	TCP
76209	45.76.173.60	172.17.112.241	TCP
76210	172.17.112.241	45.76.173.60	TCP
76211	172.17.113.25	224.0.0.251	UDP
76212	45.76.173.60	172.17.112.241	TCP
76213	172.17.113.172	224.0.0.251	UDP
76214	172.17.112.206	224.0.0.251	UDP
76215	152.42.174.114	172.17.112.241	TCP
76216	172.17.112.241	152.42.174.114	TCP
76217	152.42.174.114	172.17.112.241	TCP
76218	45.77.94.206	172.17.112.241	TCP
76219	172.17.112.241	45.77.94.206	TCP
76220	45.76.173.60	172.17.112.241	TCP

List Of Network Interface:

5. View Statistic:

```
MAIN MENU
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[9] EXPORT ALL LOGS TO FILE:
[0] EXIT

Enter your choice: 3

STATISTICS
Total Packets Captured (REAL) : 80357
Suspicious Attacks Detected   : 0
Blocked IPs                   : 0
Active Monitoring             : NO
Log File                      : ids_all_logs.txt

Press any key to continue...|
```

6. Blocked IP List :

```
MAIN MENU
[1] LIST NETWORK INTERFACES:
[2] START REAL-TIME CAPTURE:
[3] VIEW STATISTICS:
[4] VIEW BLOCKED IP LIST :
[5] VIEW ATTACK LOGS (LINKED LIST TRAVERSAL):
[6] LINEAR SEARCH IN LOGS :
[7] BINARY SEARCH (SORTED BY IP):
[8] SORTING DEMO (BUBBLE / SELECTION SORT):
[9] EXPORT ALL LOGS TO FILE:
[0] EXIT

Enter your choice: 4

BLOCKED IP LIST

No IPs are currently blocked.

Total Blocked IPs: 0

Press any key to continue...|
```

7. Attack Logs (Sorted By IP):

```
MAIN MENU
[1] LIST NETWORK INTERFACES:
[2] START REAL-TIME CAPTURE:
[3] VIEW STATISTICS:
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[5] VIEW ATTACK LOGS (LINKED LIST TRAVERSAL):
[6] LINEAR SEARCH IN LOGS :
[7] BINARY SEARCH (SORTED BY IP):
[8] SORTING DEMO (BUBBLE / SELECTION SORT):
[9] EXPORT ALL LOGS TO FILE:
[0] EXIT

Enter your choice: 5

ATTACK LOGS

No attacks detected yet.
Total Attacks: 0
Press any key to continue...
```

8.Binary Search (Sorted By IP):

```
MAIN MENU
[1] LIST NETWORK INTERFACES:
[2] START REAL-TIME CAPTURE:
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[5] VIEW ATTACK LOGS (LINKED LIST TRAVERSAL):
[6] LINEAR SEARCH IN LOGS :
[7] BINARY SEARCH (SORTED BY IP):
[8] SORTING DEMO (BUBBLE / SELECTION SORT):
[9] EXPORT ALL LOGS TO FILE:
[0] EXIT

Enter your choice: 6

LINEAR SEARCH

Enter IP address: 239.255.255.250

[NOT FOUND] IP 239.255.255.250 not found in logs.
Press any key to continue...|
```


9. Export All Log's To File:

```
MAIN MENU

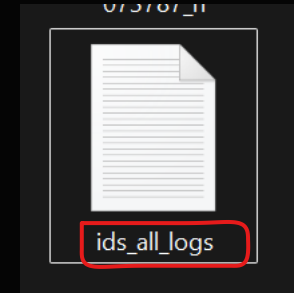
[1] LIST NETWORK INTERFACES:
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[9] EXPORT ALL LOGS TO FILE:
[0] EXIT

Enter your choice: 9

EXPORT ALL LOGS

[EXPORTED] All logs exported to ids_all_logs.txt
Total 0 attack records saved.

Press any key to continue...
```



Export Log .Text File

Source Code:



ids.cpp

Thank You